

Context Recap

Your pipeline generates trading **signals** based on **market data** + **news sentiment**.

Every trade in your `BACKTEST_RESULTS` is linked to one or more news articles (`ARTICLE_IDS`).

Now, after running the backtest, you have:

- **SIGNAL** → the model's trading decision (Buy = 1, Sell = -1, Hold = 0)
- **STRATEGY_PNL** → how much profit/loss the trade made
- **NEWS_ARTICLE** metadata → sentiment, timestamp, and topic of the articles that influenced that signal

The question becomes:

"Did positive news actually lead to profitable trades?"

"Do negative news events reliably predict downward returns?"

"Which days or articles had the most impact?"

Solution Overview

2 Aggregate by Sentiment — “How Positive or Negative News Affected Performance”

 What the output tells you:

SENTIMENT_LABEL	AVG_PNL	NUM_TRADES	Interpretation
Positive	0.0123	150	When sentiment was positive, strategy made +1.23% on average
Negative	-0.0089	130	When sentiment was negative, losses averaged -0.89%
Neutral	0.0011	70	Neutral news had little impact

Solution Overview

💡 Why this is useful:

- **Validate Signal Logic** → Confirms whether your model is interpreting sentiment correctly.
 - Example: If positive sentiment *doesn't* lead to profits, your sentiment thresholds or mapping might be wrong.
- **Feature Importance** → Shows how much sentiment contributes to profitability compared to price-based signals.
- **Tuning** → You can adjust model weights — e.g., give higher weight to strong negative sentiment if it correlates strongly with future declines.
- **Explainability** → Helps justify model behavior to non-technical stakeholders (“we trade long when news is positive because historically that led to +1.2% returns”).

Solution Overview



3 Track Article Impact Over Time — “When and How News Affected Performance”



What it does:

This analysis aggregates by **time** (usually day or hour) and checks:

- Average sentiment that day
- Total or average PnL generated
- Number of news articles influencing trades



What the output tells you:

TRADE_DAY	AVG_SENTIMENT	DAILY_PNL	Interpretation
2025-10-10	0.32	+0.08	Mostly positive news → good day for long trades
2025-10-11	-0.45	-0.10	Negative news sentiment → losing day
2025-10-12	0.05	0.00	Neutral sentiment → flat performance



Solution Overview

💡 Why this is useful:

- **Time-based Explainability** → Understand *when* news influenced markets.
 - Example: After a big product launch, news sentiment spikes → stock moves up.
- **Pattern Discovery** → Correlate market events with sentiment surges (earnings, geopolitical, product news).
- **Drift Detection** → If sentiment no longer correlates with PnL, your model might be overfitting or market behavior has changed.
- **Risk Management** → Identify “high news volatility” days where trading risk spikes.

Solution Overview